By:	AKH	Date	May 2026
Rev.--- By:	---	Date	---
Chk:		Date	

Macaulay Test

Macaulay Beam Analysis - BetaTesting

02 - Steel - Point Load at Midspan

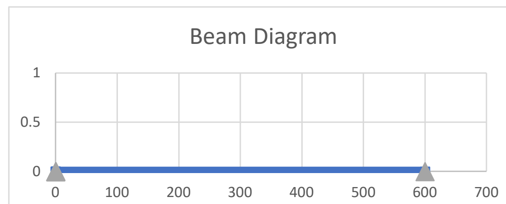
Bridge, L =		in	E =	29,000,000	lb / in^2	Length Units	in
Beam Sect =	W14x61		G =	1.740E+06	lb / in^2 = 0.06*29,000,000	Force Units	lb
A =	17.90	in^2					
I =	640.00	in^4					

Beam Segments (First Segment Begins at 0)

	Segment No.	X End	A	I	E	G
		in	in^2	in^4	lb / in^2	lb / in^2
DO NOT DELETE ROWS (GRAPHED)	1	600.00	17.90	640.00	2.900E+07	3.115E+07
	2					
	3					
	4					
	5					

Supports

	Supp. No.	X	dY (vert)	dr (rot)
			Perscribed Disp. (typ 0)	
		in	in	rad
DO NOT DELETE ROWS (GRAPHED)	1	0.0	0.0	0.000
	2	600.0	0.0	0.000
	3			
	4			
	5			

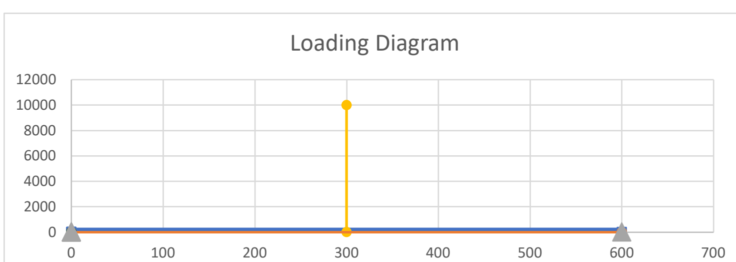


Distributed Loads (downward loads entered as NEGATIVE) - UNFACTORED

	Load No.	x1	x2	w1 (at x1)	w2 (at x2)	Load
		in	in	lb / in	lb / in	lb / ft
DO NOT DELETE ROWS (GRAPHED)	1	0.0	600.0	0.0	0.0	0.0
	2	0.0	0.0	0.0	0.0	0.0
	3	0.0	0.0	0.0	0.0	0.0
	4	0.0	0.0	0.0	0.0	0.0
	5					

Point Loads - UNFACTORED

	Load No.	X	Py	My
		in	lb	lb-in
DO NOT DELETE ROWS (GRAPHED)	1	300	-10000.00	0.00
	2	0	0.00	0.00
	3			
	4			
	5			



By:	AKH	Date	May 2026
Rev.--- By:	---	Date	---
Chk:		Date	

Macaulay Test

BEAM REACTIONS

Position (X)	nforced Disp (dy)	orced Rot (theta)	Vert Reac (Ry)	loment Ract (Mz)	Note
(L)	(L)	(rad)	(F)	(F-L)	(---)
0.00	0.0	0.0	5,000.0	0.0	Support reaction
600.00	0.00	0.00	5,000.0	0.0	Support reaction

Vert Reac
(kip)
5.000
5.000

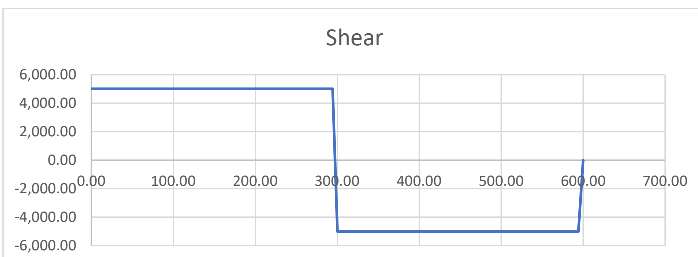
BEAM REACTIONS EQUILIBRUM CHECK

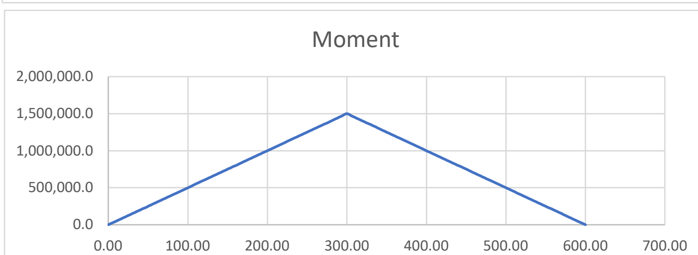
Item	Units	Loads	Reactions	Residual	Check Sum
Sum Fy	F	-10,000.0	10,000.0	0.0	OK
Sum M @ x=0	F-L	-3.0000E+06	3.0000E+06	0.0	OK

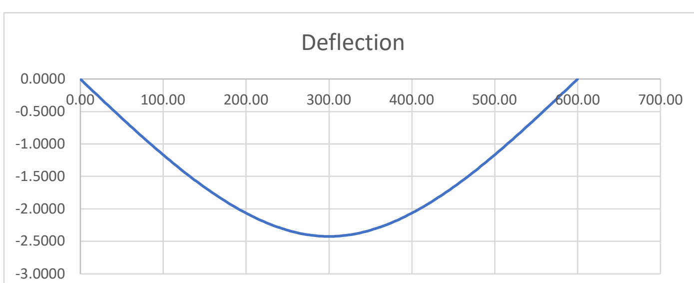
BEAM MAX/MIN LOAD EFFECTS

Item	Position (X)	Value
(---)	(L)	(F or F-L)
Max +V	0.00	5,000.0
Max -V	300.00	-5,000.0
Max +M	300.00	1,500,000.0
Max -M	0.00	0.0
Max Defl	300.00	-2.424

Value	
(---)	
5.00	kip
-5.00	kip
125.00	kip-ft
0.00	kip-ft
-2.424	in

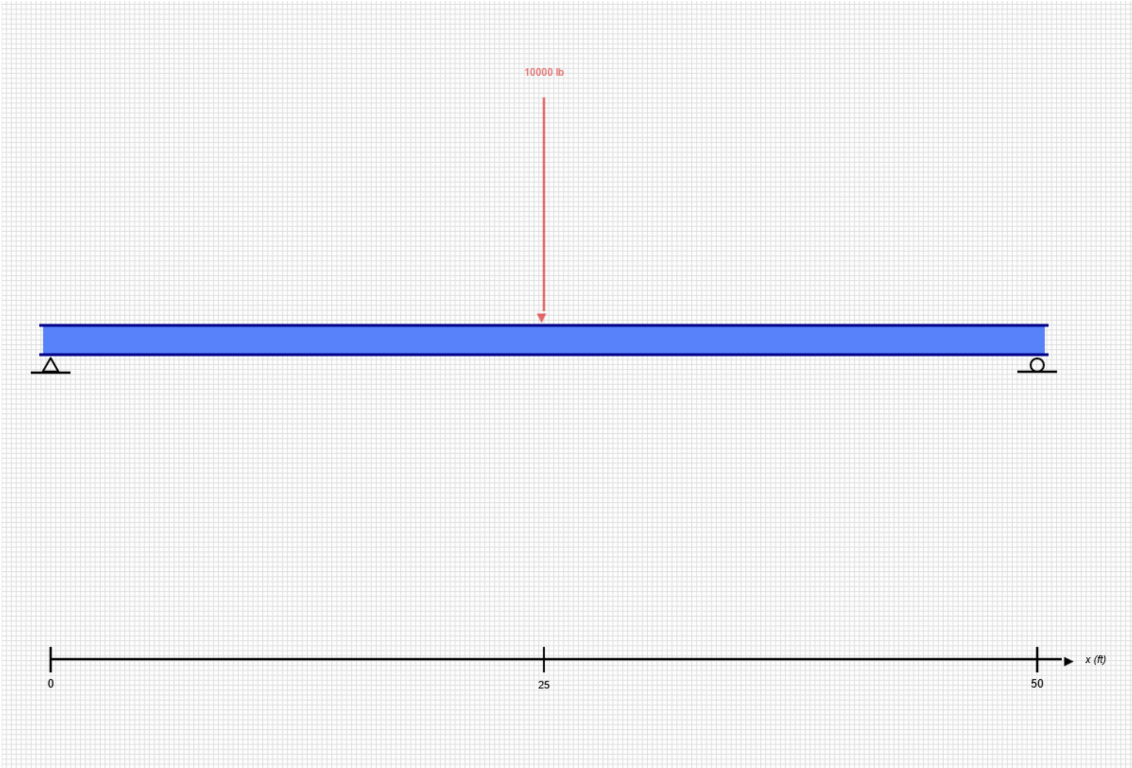






SKYCIV BEAM ANALYSIS REPORT

Load Combination: DL



Software: SkyCiv Beam v3.3.2
Tue Jun 09 2026 12:42:55 GMT-0400 (Eastern Daylight Time)

Project Info

File Name: 01-Steel-Beam

Included in this Report:

- Input Summary
- Beam Section
- Free Body Diagram (FBD)
- Analysis Summary
- Analysis Results

INPUT SUMMARY

General Info

Beam Length:	50 ft
Section Name:	W14x61
Self Weight:	False

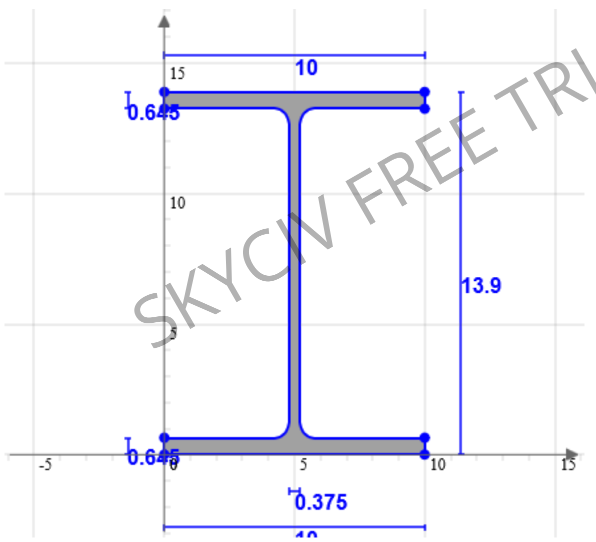
Supports

Support Type	Location
Pinned	0 ft
Roller	50 ft

Loads

Load Type	Location	Magnitude	Load Case
Point Load	25 ft	-10000 lb	DL

Beam Section



Geometric Properties		
A	17.9	in^2
C_z	5	in
C_y	6.95	in

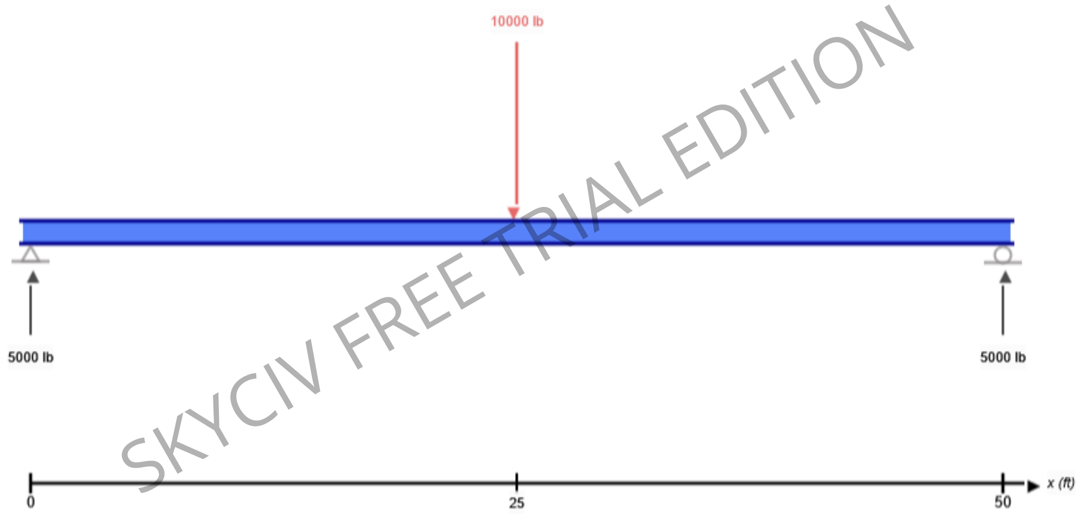
Bending Properties		
I_z	640	in^4
I_y	107	in^4

Shear Properties		
A_z	11.277	in^2
A_y	4.99	in^2

Torsion Properties		
J	2.19	in^4
r	0.991	in

Shape	Material	E (ksi)	v	ρ (lb/ft^3)
W14x61	Structural Steel	29000	0.27	490

FREE BODY DIAGRAM



RESULT SUMMARY

Check	Status	Limit	Ratio	Max
Deflection	FAIL	L/250	1.012	L/247
Custom Stress Limit	PASS	39 ksi	0.418	16.289 ksi
Material Yield	PASS	36 ksi	0.452	16.289 ksi
Material Strength	PASS	58 ksi	0.281	16.289 ksi

ANALYSIS RESULTS

Reactions

Support at	R_x (lb)	R_y (lb)	M (kip-ft)
0	0	5000	0
50	0	5000	0

Force Extremes

Result	Max	Min
Bending Moment	125 kip-ft	0 kip-ft
Shear	5000 lb	-5000 lb
Displacement Y	0 in	-2.425 in

Stress Extremes

Result	Max (ksi)	Min (ksi)
Bending Stress	16.289	-16.289
Shear Stress Total	1.068	1.068
Max Combined Normal Stress	16.289	0
Min Combined Normal Stress	0	-16.289